

STEVEN (CHENG-JU) WU

(669) 288-4917 | cw3834@columbia.edu | linkedin.com/in/steven-cj-wu/

EDUCATION

Columbia University Master of Science in Mechanical Engineering Relevant Course: Robotics Studio, Continuous Control Systems, Microelectromechanical Systems.	New York, NY Expected Dec 2027
Santa Clara University (SCU) Bachelor of Science in Mechanical Engineering Relevant Courses: Composite Materials, Fluid Mechanics, Finite Element Theory and Applications, Engineering and Design with Applied Machine Learning, Computational Mechanics, Continuum Mechanics.	Santa Clara, CA Jun 2026

WORK EXPERIENCE

Santa Clara University GIFTED Lab Undergraduate Research Assistant • Awarded the SCU Kuehler Undergraduate Research Grant to conduct research on machine-learning-driven topology optimization. • Developed a PINN-based topology optimization framework for 60 × 20-element structural domains, integrating minimum feature-size constraints to generate additive-manufacturing-compatible designs. • Implemented a multi-network architecture coupling material-density prediction with displacement-field estimation, reducing optimization runtime by 5% versus SIMP-based methods in complex solutions. • Built and trained neural network models in PyTorch using automatic differentiation; analyzed and visualized material-density evolution with NumPy and Matplotlib.	Santa Clara, CA Jun 2025 - Jan 2026
SINBON Electronics Co., Ltd R&D Mechanical Engineering Intern • Built and iterated injection-molded plastic and sheet-metal components in Creo for OEM electronics assemblies, considering manufacturability, structural requirements, tolerances, assembly interfaces, and cost. • Produced 3D CAD models, engineering drawings, BOMs, and design documentation to facilitate prototyping, supplier sourcing, assembly, and production. • Collaborated with electrical and manufacturing engineers to resolve packaging, tolerance, and DFM issues across 3 product designs, managing prototype-to-production development.	New Taipei City, TWN Jul 2024 - Sep 2024

ACTIVITIES

SCU Bronco Air Design Build Vertical Flight (DBVF) Pegasus • Created and fabricated an autonomous hexacopter for the Vertical Flight Society DBVF Competition, placing 9th overall. • Designed airframe and motor-arm assemblies in SOLIDWORKS, reducing structural weight by 47.4% through FEA-driven topology optimization while maintaining FOS > 2.0 and deflection < 1 mm. • Integrated Raspberry Pi and CubePilot Cube Orange+ into a 6-motor autonomous flight system, supporting waypoint navigation and mission-route planning.	Jun 2025 - Jun 2026
SCU Robotics System Lab Vertical Profiler • Diagnosed post-deployment damage on 3 vertical profilers and reverse-engineered replacement components in SOLIDWORKS for 3D printing. • Designed modular SOLIDWORKS mounts for action cameras and sensors, enabling rapid field reconfiguration across multiple instrumentation setups.	Jun 2025 - Jun 2026
SCU Bronco Racing Formula SAE Electric • Created side-impact structures in SOLIDWORKS for a Formula SAE Electric vehicle, meeting structural, cockpit-clearance, and driver-safety requirements. • Performed FEA on cockpit and side-impact structures under simulated 70 km/h side-impact and 4.5G worst-case loading, comparing material configurations for structural integrity and weight reduction.	Sep 2022 - Jun 2024

LANGUAGE AND IT SKILLS

- Languages: Mandarin (Native), German (Elementary).
- Technical: SOLIDWORKS, Creo, MATLAB, 3D Printing, Laser Cutting, Pytorch, Python, C++, GD&T.